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SUPERSEDING
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FEDERAL SPECIFICATION

COMPONENTS FOR INTERIOR ALARM SYSTEMS

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers intrusion detection equipment which when deployed as an intrusion detection system, is designed to include, but not be limited to detection of intrusions and to assist in protection of national security information, sensitive information and equipment, high dollar items, and weapons. The intrusion detection equipment, when selectively assembled, provides interior detection capabilities specifically designed for security application.

1.2 Classification.

1.2.1 Units and types. Intrusion detection system components shall consist of at least one of the following units:

Unit 1 - Detector (see 1.2.2 and 3.5)
Unit 2 - Annunciator (see 3.6 through 3.6.4)
Unit 3 - Signaling Technology (see 1.2.3 and 3.7)
Unit 4 - Control Unit (see 3.8)

1.2.2 Detectors (Unit 1). Detectors furnished under this specification shall be one of the following types:

Type I - Balanced Magnetic Switch
Type II - Foil
Type III - Breakwire
Type IV - Light Threshold motion detector
Type V - Infrared Beambreak detector
Type VI - Passive Infrared (PIR) Detector
Type VII - Vibration Detector
Type VIII - Proximity Detector
Type IX - Ultrasonic Motion Detector
Type X - Microwave Motion Detector
Type XI - Pressure Mat Detector
Type XII - Closed Circuit Television (CCTV) Motion Detector

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Type XIII - Glass Breakage Detector
Type XIV - Flexible Cable Detector
Type XV - Infrasonic
Type XVI - Sound Discriminator (Sonic)

1.2.3 Circuit supervision. The supervised alarm transmission means shall be of the classes specified in 3.7 and 3.7.1.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on the date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601 - Boxes, Wood, Crated-Plywood

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of the Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

(Single copies of the specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in the agencies.)

Military Standards:

MIL-STD-105 Sampling Procedures and Tables for

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Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

MIL-STD-461 Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 Electromagnetic Interference Characteristics, Measurements of.

MIL-STD-810 Environmental Test Methods and Engineering Guidelines.

(Copies of military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials.

ASTM D3951 Standard Practices for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Material, 1916 Race St., Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General. The annunciator units, control units, and the line supervision under this specification shall be products that meet the minimum specifications required herein and that have been tested and passed the tests specified herein. The detector units offered under this specification shall be units which are qualified for listing on the applicable qualified products list.

3.2 Parts and materials. Parts and materials for components shall be as specified herein. Those not specified shall be equivalent to and interchangeable with the corresponding parts, materials or processes in the manufacturer's normal commercial product and shall be subject to all provisions of this specification. Normal commercial product shall be interpreted to mean an end item acceptable under this specification.

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3.2.1 Custom tools. Any custom tools required for the adjustment or maintenance of products specified in this specification shall be provided with each unit shipped.

3.3 Equipment enclosures. All enclosures for equipment supplied under this specification shall be protected against tampering. Enclosures shall be equipped with tamper switches or triggering mechanisms electrically compatible with the alarm system. Such tamper switches or trigger mechanisms shall be capable of being disabled when the enclosure is opened for service. Tamper switches shall be armed automatically when the enclosure is reclosed. Internal wiring of equipment shall be such that the tamper switches and triggering mechanisms are not bypassed though the detector itself is operating in the "ACCESS" mode. All controls that affect the sensitivity of the units shall be located inside the tamper resistant enclosure. If the contract specifies that the enclosures be metal they shall be constructed of sheet steel and shall be not less than No. 16 U.S. Standard Gauge. Enclosures shall not have prepunched knockouts. Where doors are mounted on hinges with exposed pins, the hinges shall be tack welded to prevent easy removal. The tamper switches shall be located so that activation occurs when the door or cover is moved as little as 5 mm (0.2 inches) from its normally closed position (unless otherwise specified). Movable covers and doors shall have flange edges at least 5 mm (0.2 inches) in width and shall be equipped with high security locks at the latch edge.

3.4 Electronic components. To the extent practical all electronic components shall be the solid state type.

3.5 Detectors. All detectors shall initiate an alarm signal under any of the following conditions: (1) When sensing a stimulus or condition for which it was designed to react; (2) primary power fails and secondary power does not take over properly; (3) a tamper switch or triggering mechanism is activated. Terminals shall be located within the detector enclosure and shall be readily accessible to permit wiring connections to be made.

3.6 Annunciator units. Annunciator units shall be so designed that when connected with the ancillaries into a detection circuit they provide the means to remotely monitor the condition and control the operation of the detection circuit. A tamper alarm shall be generated when the line between the annunciator and the detector is opened, shorted, or grounded. Annunciator units shall be electrically compatible with the detector and circuit supervision equipment described herein and shall be of modular design capable of being installed with other annunciators in a rack, console, or cabinet. When specified (see

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6.2), individual annunciator units shall be furnished in appropriate enclosures. To the extent practicable, equipment related to the annunciator such as standby-battery, power supply, battery charging equipment, audible alarm, and circuit supervisor functions, shall be contained in the same enclosure.

3.6.1 Annunciator access/secure, and reset switches. The annunciator panel may have an access/secure switch and shall have an alarm resets switch. An alarm shall create a lock-on condition to be displayed by both audible and visual means, which shall require a manual restoration and controls shall be provided to reset the system. A received alarm shall cause the visual device to operate in a pulsing or flashing mode, or other visual indication. The annunciator shall have a means of silencing the audible signal from a particular zone during prolonged alarm conditions. However, the visible signal shall remain illuminated on the annunciator panel or CRT display until the system is restored to normal operation, except that the visual signal shall be changed from pulsing/flashing to a steady mode. CRT displays will continue to show an alarm until all areas are secured. The silencing control shall be so connected that the audible alarm signal shall be activated upon receipt of an alarm from another zone. When a detection circuit is conditioned for authorized entry into the protected area (Access Mode), annunciators shall continue to indicate alarms if circuit supervision limits are exceeded or any tamper switches are disturbed.

3.6.2 Annunciator readout. The annunciator shall be equipped with a readout of the type specified in 3.6.2.1 or of the type specified in 3.6.2.2.

3.6.2.1 Electronic type. This type of readout shall have either a Cathode Ray Tube (CRT) (MTBF of 40,000 hours or greater), or a light emitting diode (LED) (MTBF of 50,000 hours or greater) or liquid-crystal display (LCD) type of display and a printer, if used, for recording alarm conditions. The display, as a minimum, shall display any location that has a change of status and show the present status of that location. The printer, if used, shall printout a location that has a change of status, the present status of that location, and the date and time of day the change occurred.

3.6.2.2 Colored signal lights type. This type shall indicate, by duplicate colored signal lights, (MTBF of 20,000 hours or greater), the following conditions:

(1) That the detector(s) and circuits are in the secure condition (green light);

(2) That a detector in the protected area is in an alarm

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condition, or that circuit supervisor limits have been exceeded. If the detector provides an indication of component failure, this also should be annunciated (red lights);

(3) That the system is operating in an access condition (yellow amber lights);

(4) That the detector(s) in the protected area is operating on standby power (white lights).

In addition to the visual signal lights, an audible signal shall be annunciated whenever the system changes from one condition to another. However, the audible signaling device need not be part of each individual module.

3.6.3 Annunciator construction. Individual annunciator modules of a given manufacturer shall be interchangeable to facilitate maintenance. Plug and sockets shall be used to the extent possible. All parts of the annunciator shall be easily identified and readily accessible to authorized maintenance personnel. All controls required for normal operation shall be permanently and conspicuously marked. All controls not required for operation of the system shall not be readily accessible to the operator when the equipment is installed in a rack or console.

3.6.4 Annunciator connections. The annunciator shall be provided with terminals for all required external connections.

3.7 Circuit supervision units. The circuit supervisors shall provide security to the communication link between the detector and the annunciator. The circuit supervisors shall be classed according to relative security as specified herein.

3.7.1 Line supervision classes. The circuit supervisors shall provide security to the communication link between the individual control units in the protected areas and the monitor display. The level of security provided to the data information on the communication lines between the protected areas and the monitor are designated as follows:

Class A - Compromise would require the application of national level computer analysis before the attempt to attack. A collection of encrypted data from the communication lines to be compromised must be available for the analysis. Special equipment or software is necessary and it would require professional engineering skills and experience to implement.

Class B - Compromise would require the use of a microprocessor and prior computer analysis. The compromise will require

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significant competence of the technical personnel and working samples of the equipment.

Class C - Compromise would require the use of special logic circuitry, without requiring the use of prior computer analysis.

Class D - Compromise would require the use of active analog circuitry including transistors and tape recorders.

Class E - Compromise would require the application of a combination of passive or active components such as filters and phase shifters.

Class F - Compromise would require only the insertion of a single passive or active component.

3.8 Access control units. The premises control unit shall be so designed that when operated it permits access to a protected area without activating an alarm signal. The unit shall consist of circuitry installed in a metal enclosure, the cover of which shall contain a two or more position, key-operated switch. The positions shall be labeled ACCESS and SECURE plus whatever other labels are required to denote the functions designed into the unit submitted for qualification. Turning the switch from SECURE to ACCESS shall alter the signal(s) to the annunciator and shall deactivate the detection device, however the tamper switches shall continue to be monitored. Turning the switch from ACCESS to SECURE shall alter the signal(s) to the annunciator and shall activate the detection device to enable monitoring of alarm and tamper signals.

3.9 Primary power requirements. The alarm systems shall operate on 115 or 220 volts (±10%), as specified, and shall operate on either 50 or 60 Hz (± 5%) or as specified (see 6.2).

3.9.1 Standby-battery. A standby-battery source for use in the event the primary power source fails shall be provided. The standby-battery power source, when fully charged, shall be capable of maintaining full operation of the alarm system for not less than four (4) continuous hours at temperatures from minus 30°C (22° Fahrenheit) to plus 49°C (+120° Fahrenheit). Switch-over to battery power shall be instantaneous and automatic upon failure or restoration of the primary power source and shall not create alarms on annunciator modules. However, loss of primary power shall be noted by separate indication.

3.9.2 Batteries. Batteries shall be float charged and so arranged that they are fully charged at all times when primary power is available. Charger shall be of ample capacity to recharge the batteries from a fully discharged state to not less

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than 85% of capacity within 24 hours. Controls shall provide monitoring to preclude overcharging, or the circuit and battery combination shall intrinsically provide the feature.

3.9.3 Ventilation. Power supplies shall be vented or otherwise protected against excessive heat.

3.10 Identification markings. Each major component furnished under contract or order shall have the manufacturer's name and address, unit model and serial number, year manufactured, and the symbol, W-A-450, indented or embossed on a metal plate that shall be firmly affixed in such a place to prevent identification of the unit without removing the cover.

3.11 Technical manuals and operator instructions. Manufacturer's installation recommendations/instructions, applications guides, equipment limitations, specifications for operation as listed in section 3, technical manuals, and operator and maintenance instructions, shall be complete and comprehensive and delivered with the equipment, as specified (see 6.2). The manual shall include schematics and wiring diagrams for all components and complete electrical values and ratings as well as a section containing the theory of operation of all portions of the total system, i.e., specific technical descriptions relating to respective circuit schematics. All production run changes or component modifications, and revisions to technical information, shall be reported to the recipient installation security office for 36 months following date of system acceptance by that security office.

3.12 Workmanship. All components shall be manufactured and finished in such a manner as to meet specification requirements. Circuit wiring shall be neat with good electrical connections that are mechanically secure. Components and parts shall be free from characteristics and defects that might affect operation or serviceability.

3.13 Spare parts. Spare parts, as required by the procuring activity, shall be furnished as specified (see 6.2).

3.14 Electromagnetic radiation interference. Alarm system units shall be designed to minimize susceptibility to external electromagnetic fields and to minimize adverse effects on electronic equipment in the the unit's vicinity, in accordance with the latest revision of MIL-STD-461 and MIL-STD-462. Equipment shall comply with all relevant FCC regulations pertaining to licensed or unlicensed transmitters, as appropriate.

3.15 Lightning protection. The annunciator and ancillary

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devices shall be protected against lightning-induced power surges that would damage the equipment. The protection device shall be of unit construction and shall not include fusible elements that will require replacement.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. Except as otherwise specified herein, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own or any other inspection facility or services acceptable to the Government. Inspection records of the examinations and tests shall be maintained by the manufacturer for at least one year, and shall be available for Government inspection. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Components and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and material used are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.3 Inspection for acceptance. Units to be offered for acceptance on the GSA schedule may be inspected by a Government Inspector at any time during manufacturing processes to assure that the units function as intended, that parts and material are complete and as specified, that parts and components are properly located, that mechanical and electrical connections are secure, that soldered connections provide good electrical conductivity, that component enclosures are protected against tampering, and that all workmanship details are of good quality. When specified (see 6.2), acceptance inspection shall be performed at the destination after equipment is installed and ready for use.

4.3.1 Testing for acceptance. To ensure continued compliance with specification requirements relating to operation performance, the Government reserves the right to purchase from the manufacturer's regular production samples of alarm system components for testing. The testing shall be performed by a facility designated by the Government and shall be at no cost to the manufacturer. Failure to meet testing requirements shall provide reason to suspend GSA listing of the manufacturer's product. Once the testing requirements have again been successfully met as per this document, the manufacturer must resubmit an application to be placed on the GSA schedule again.

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4.4 Inspection of preparation for delivery. An inspection of preparation for delivery shall be made in accordance with ASTM D3951.

4.5 Qualification. In accordance with 3.1, products submitted for qualification shall be inspected for workmanship as specified in 4.3 and subjected to the tests in the individual product specifications. Failure to meet inspection or testing requirements shall be considered as failure to meet requirements for GSA listing.

4.5.1 Testing agency. Qualification testing shall be performed by an independent laboratory of the manufacturer's choice, at his expense. Requests for authorization to submit products for qualification testing shall be in accordance with the Provisions Governing Qualification of Federal Property Management Regulation (FPMR) 101-29.

4.5.2 Testing costs. All costs entailed in the qualification testing of proposed products shall be borne by the supplier.

4.5.3 Test samples. The samples submitted for qualification testing shall consist of all components required by this specification and individual product specifications, necessary for a complete working unit of the type the supplier proposes to furnish. Included with the product sahll be a complete set of technical manuals, wiring diagrams and schematics for use during testing.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with ASTM D3951. The level of preservation and packaging shall be A or C, and the level of packing shall be A, B, or C, as specified (see 6.2).

5.2 Civil agency marking. In addition to any special marking (see 6.2) required by the contract or order, the interior packages and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Intrusion detection systems constructed with components qualified under this specification are intended to notify protection personnel (e.g., response force) whose responsibility is the protection of areas requiring special

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security measures. Properly assembled systems are highly resistant to neutralization or compromise. Guidance in selecting a particular item covered by this specification for a particular situation is not intended and is beyond the scope of this document.

6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the supplier is called to this requirement, and manufacturers are urged to arrange to have the product they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity reponsible for the qualified products list is the General Services Administration, Federal Supply Service, Furniture Commodity Center, Engineering Division, Washington, DC 20406, and information pertaining to qualification of products may be obtained from that activity.

6.3 Ordering data. Purchaser should exercise any desired options permitted herein and include the following information in procurement documents:

(a) Title, symbol and date of this specification.
(b) Type of detector unit and number required (see 1.2.2)
(c) Number of annunciators required (see 1.2.1)
(d) Type of annunciator readout required (see 3.6.2)
(e) Enclosure for single annunciator required (see 3.6)
(f) Class of circuit supervision unit required (see 1.2.3 and 3.7)
(g) Number of premise controls desired (see 3.6)
(h) Power requirements (see 3.9)
(i) Spare parts required (see 3.3)
(j) Destination inspection required (see 4.3)
(k) Level of packaging and packing required (see 5.1)
(l) Whether special marking required (see 5.1.1)

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